



LocoNet Opcode Summary

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Checksum

The data bytes on LocoNet® are defined as 8 bit data with the most significant bit (transmitted last in the 8 bit octet) as an OPCODE flag bit. If the MS bit , D7, is 1 the 7 least significant bits are interpreted as a network OPCODE . The opcode byte may only occur once in a valid message and is the FIRST byte of a message. All the remaining bytes in the message must have a most significant bit of 0 , including the last CHECKSUM byte. The CHECKSUM is the 1's COMPLEMENT of the byte wise Exclusive Or of all the bytes in the message, except the CHECKSUM itself. To validate data accuracy, all the bytes in a correctly formatted message are Exclusive Or'ed. If this resulting byte value is "FF" hexadecimal, the message data is accepted as good.

Message Length

The OPCODES may be examined to determine message length and if subsequent response message is required. Data bits D6 and D5 encode the message length. **D3=1 implies Follow-on message/reply:**

D7	D6	D5	D4	D3	D2	D1	D0	
1	0	0	F	D	C	B	A	Message is 2 bytes, including Checksum
1	0	1	F	D	C	B	A	Message is 4 bytes, inc. checksum
1	1	0	F	D	C	B	A	Message is 6 bytes, inc checksum
1	1	1	F	D	C	B	A	Message in N bytes, where next byte in message is a 7 bit BYTE COUNT.

The A,B,C,D,F are bits available to encode 32 OPCODES per message length.

Refresh Slots

The model of the MASTER refresh stack is an array of up to 120 read/write refresh SLOTS. The slot address is a principal component and is generally the second byte or 1st argument of a message to the master. The refresh SLOT contains up to 10 data bytes relating to a Locomotive and also controls a task in the Track DCC refresh stack. Most mobile decoder or Locomotive operations process the SLOT associated with the Locomotive to be controlled. The SLOT number is a similar shorthand ID# to a "file handle" used to mark and process files in a DOS PC environment. Slot addresses 120-127 ARE reserved for System and Master control. Slot #124 (\$7C) is allocated for read/write access to the DCS100 programming track, and the format of the 10 data bytes is not the same as a "normal" slot. See later.

Standard Address Selection

To request a MOBILE or LOCOMOTIVE decoder task in the refresh stack, a Throttle device requests a LOCOMOTIVE address for use,(opcode <BF>,<loco adr hi>,<loco adr lo> , <chk>). The Master (or PC in a Limited Master environment) responds with a SLOT DATA READ for the SLOT ,(opcode <E7>,) ,that **contains** this Locomotive address and all of its state information. If the address is currently not in any SLOT, the master will load this NEW locomotive address into a new SLOT ,[speed=0, FWD, Lite/Functions OFF and 128 step mode]and return this as a SLOT DATA READ. If no inactive slots are free to load the NEW locomotive address, the response will be the Long Acknowledgment ,(opcode <B4>,) , with a "fail" code, 0. Note that regular "SHORT" 7 bit NMRA addresses are denoted by <loco-adr hi>=0. The Analog , Zero stretched, loco is selected when both <loco adr hi>=<loco adr lo>=0. <Loco adr lo> is always a 7 bit value. If <loco adr hi> is non-zero then the Master will generate NMRA type 14 bit or "LONG" address packets using all 14 bits from <loco adr hi> and <loco adr lo> with Loco adr Hi being the MOST significant address bits. Note that a DT200 Master does NOT process 14 bit adr requests and will consider the <loco adr hi> to always zero. You can check the <TRK> return bits to see if the Master is a DT200.

The throttle must then examine the SLOT READ DATA bytes to work out how to process the Master response. If the STATUS1 byte shows the SLOT to be COMMON, IDLE or NEW the throttle may change the SLOT to IN_USE by performing a **NULL MOVE** instruction ,(opcode <BA>,<slotX>,<slotX>,<chk>) on this SLOT. **This activation mechanism is used to guarantee proper SLOT usage interlocking in a multi-user asynchronous environment.**

If the SLOT return information shows the Locomotive requested is IN_USE or UP-CONSISTED (i.e. the SL_CONUP, bit 6 of STATUS1 =1) the user should NOT use the SLOT. Any UP_CONSISTED locos must be UNLINKED before usage! Always process the result from the LINK and UNLINK commands, since the Master reserves the right to change the reply slot number and can reject the linking tasks under several circumstances. Verify the reply slot # and the Link UP/DN bits in STAT1 are as you expected.

The throttle will then be able to update Speed./Direction and Function information . Whenever SLOT information is changed in an active slot , the SLOT is flagged to be updated as the next DCC packet sent to the track. If the SLOT is part of linked CONSIST SLOTS the whole CONSIST chain is updated consecutively.

If a throttle is disconnected from the LocoNet®, upon reconnection (if the throttle retains the SLOT state from before disconnection) it will request the full status of the SLOT it was previously using. If the reported STATUS and Speed/Function data etc., from the master exactly matches the remembered SLOT state the throttle will continue using the SLOT. If the SLOT data does not match, the throttle will assume the SLOT was purged free by the system and will go through the setup “log on” procedure again.

With this procedure the throttle does not need to have a unique “ID number”. SLOT addresses DO NOT imply they contain any particular LOCOMOTIVE address. The system can be mapped such that the SLOT address matches the LOCOMOTIVE address within, if the user directly Reads and Writes to SLOTS without using the Master to allocate Locomotive addresses

Dispatching

Active Locomotives (including Consist TOP) SLOTS may be released for assignment to BT2 throttles in the “DISPATCH” mode. In this case a BT2 operating in its normal mode will request a DISPATCH SLOT that has been prepared by a supervisor type device. This is included for Club type operations where simpler throttles with limited capabilities are given to Engineers (Operators) by the Hostler or Dispatcher.

To **DISPATCH PUT** a slot , perform a SLOT MOVE to Slot 0. In this case the Destination Slot 0 is not copied to, but the source SLOT number is marked by the system as the DISPATCH slot. This is only a “one deep stack”.

To **DISPATCH GET**, perform a SLOT MOVE from Slot 0 (no destination needed). If there is a DISPATCH marked slot in the system, a SLOT DATA READ (<E7>,,,) with the SLOT information will be the response. If there is NO DISPATCH slot, the response will be a LONG ACK (opc <B4>,,) with the Fail code,00.

Future Expansion Codes

Immediate codes may be sent to the Master by a device. These are converted to DCC packets and sent as the next packet to the rails. They are not entered into any refresh stack. These are available in a system based on the DCS100/“Chief”.

Opcodes for access to an auxiliary Service mode Programming Track are included. These requests are not entered in the main DCC packet stream .

Note that several confusing expansions and opcode sequences have been stripped from this LocoNet® version. An experimenter who implements this protocol correctly should have no problems running on a LocoNet® that has other expanded features. Again, we recommend resisting the temptation to “optimise” or take shortcuts with this protocol since it will lead to guaranteed future problems with your hardware and software.

2 Byte Message Opcodes

FORMAT

<OPC> , <CKSUM>

Opcode		Response		
Symbol	Code	Description	Success	Fail
OPC_IDLE	0x85	FORCE IDLE state, B'cast emerg. STOP	-	-
OPC_GPON	0x83	GLOBAL power ON request	-	-
OPC_GPOFF	0x82	GLOBAL power OFF req	-	-
OPC_BUSY	0x81	MASTER busy code, NUL	-	-

4 Byte Message Opcodes

FORMAT

<OPC> , <ARG1> , <ARG2> , <CKSUM>

Opcode		Description	Arguments		Response	
Symbol	Code		Arg1	Arg2	Success	Fail
OPC_LOCO_ADR	0xBF	Request loco address, if not found master puts address in a free slot.	0	ADR	OPC_SL_RD_DATA	OPC_LONG_ACK No free slot available. ACK1=0
OPC_SW_ACK	0xBD	Request switch with acknowledge function.	SW1	SW2	OPC_LONG_ACK, ACK1=0x7F	OPC_LONG_ACK, ACK1=0x00
OPC_SW_STATE	0xBC	Request state of switch.	SW1	SW2	OPC_LONG_ACK, ACK1=0x7F	OPC_LONG_ACK, ACK1=0x00
OPC_RQ_SL_DATA	0xBB	Request slot data/status block	SLOT	0	OPC_SL_RD_DATA	
OPC_MOVE_SLOTS	0xBA	Move slot SRC to DST	SRC	DST	OPC_SL_RD_DATA	
OPC_LINK_SLOTS	0xB9	Link slot ARG1 to slot ARG2	SL1	SL2	OPC_SL_RD_DATA	
OPC_UNLINK_SLOTS	0xB8	Unlink slot ARG1 from slot ARG2	SL1	SL2	OPC_SL_RD_DATA	
OPC_CONSIST_FUNC	0xB6	Set FUNC bits in a CONSIST uplink element	SLOT	DIRF		

OPC_SLOT_STAT1	0xB5	Write slot stat1	SLOT	STAT1		
OPC_LONG_ACK	0xB4	Long acknowledge	LOPC	ACK1		
OPC_INPUT_REP	0xB2	General sensor input codes	IN1	IN2		
OPC_SW_REP	0xB1	Turnout sensor state report	SN1	SN2		
OPC_SW_REQ	0xB0	Request switch function	SW1	SW2		OPC_LONG_ACK, LOPC=0x30, ACK1=0x00
OPC_LOCO_SND	0xA2	Set slot sound function	SLOT	SND		
OPC_LOCO_DIRF	0xA1	Set slot direction, function 0-4 state	SLOT	DIRF		
OPC_LOCO_SPD	0xA0	Set slot speed	SLOT	SPD		

6 Byte Message Opcodes

FORMAT

<OPC>, <ARG1>, <ARG2>, <ARG3>, <ARG4>, <CKSUM>

Symbol	Code	Description	Arg1	Arg2	Arg3	Arg4	Success	Fail
OPC_MULTI_SENSE	0xD0	power management and transponding	type	zone and section	addr	addr		
OPC_UHLI_FUN	0xD4	Function 9-28 by Uhlenbrock	0x20	slot	function group	function		

Variable Byte Message Opcodes

FORMAT

<OPC>, <COUNT>, <ARG1>, <ARG2>, <ARG3>, . . . , <ARG(COUNT-3)>, <CKSUM>

Opcode			Arguments											
Symbol	Code	Description	Count	Arg1	Arg2	Arg3	Arg4	Arg5	Arg6	Arg7	Arg8	Arg9	Arg10	Arg1
OPC_WR_SL_DATA	0xEF	Write slot data.	0x0E	SLOT#	STAT1	ADR	SPD	DIRF	TRK	SS2	ADR2	SND	ID1	ID2
OPC_WR_SL_DATA	0xEF	Write PT slot data.	0x0E	0x7C	PCMD	0	HOPSA	LOPSA	TRK	CVH	CVL	DATA7	0	0
OPC_WR_SL_DATA	0xEF	Write Fast Clock slot data.	0x0E	0x7B	CLK RATE	FRAC MINSL	FRAC MINSH	256-MINS 60	TRK	256-HRS 24	DAYS	CLK CNTR	ID1	ID2
OPC_SL_RD_DATA	0xE7	Slot data response.	0x0E	SLOT#	STAT1	ADR	SPD	DIRF	TRK	SS2	ADR2	SND	ID1	ID2
-	0xE6	Programming service mode aborted.	0x10/0x15											

Opcode			Arguments														
Symbol	Code	Description	Count	Arg1	Arg2	Arg3	Arg4	Arg5	Arg6	Arg7	Arg8	Arg9	Arg10	Arg11	Arg12	Arg13	Response
OPC_PEER_XFER	0xE5	Move 8 bytes peer to peer, SRC→DST.	0x10	SRC	DSTL	DSTH	PXCT1	D1	D2	D3	D4	PXCT2	D5	D6	D7	D8	-

Opcode			Arguments							
Symbol	Code	Description	Count	Arg1	Arg2	Arg3	Arg4	Arg5	Arg6	Response
OPC_LISSY_REP	0xE4	Lissy IR report	0x08	0x00	high unit, direction	low unit	high addr	low addr	-	-
OPC_WHEELCNT_REP	0xE4	Wheel counter report	0x08	0x40	high unit, direction	low unit	high count	low count	-	-

Opcode			Arguments											
Symbol	Code	Description	Count	Arg1	Arg2	Arg3	Arg4	Arg5	Arg6	Arg7	Arg8	Arg9	Response	
OPC_LISSY_REP	0xE4	RFID-5 report	0x0C	0x41	addr. high	addr. low	RFID-0	RFID-1	RFID-2	RFID-3	RFID-4	RFID-HI	-	

Opcode			Arguments												
Symbol	Code	Description	Count	Arg1	Arg2	Arg3	Arg4	Arg5	Arg6	Arg7	Arg8	Arg9	Arg10	Arg11	Response
OPC_LISSY_REP	0xE4	RFID-7 report	0x0E	0x41	addr. high	addr. low	RFID-0	RFID-1	RFID-2	RFID-3	RFID-4	RFID-5	RFID-6	RFID-HI	-

Opcode			Arguments									
Symbol	Code	Description	Count	Arg1	Arg2	Arg3	Arg4	Arg5	Arg6	Arg7	Arg8	Response
OPC_IMM_PACKET ¹⁾	0xED	Send n-byte packet immediate.	0x0B	0x7F	REPS	DH1	IM1	IM2	IM3	IM4	IM5	OPC_LONG_ACK ACK1=0x7F if not limited ACK1=lim if limited ACK1=0x00 If busy

¹⁾ The SLOT DATA bytes are, in order of TRANSMISSION for <E7> READ or <EF> WRITE NOTE SLOT 0 <E7> read will return MASTER config information bytes .